# COMMERCIAL ITEM DESCRIPTION

## CUTTER-VACUUM, ORTHOPEDIC CAST (110/230 V, 50/60 Hz, AC)

The General Services Administration has authorized the use of this commercial item description

This commercial item description covers an electric motor driven saw used to cut orthopedic casts on the body without injuring the patient combined with a portable vacuum cleaner used to dispose of plaster and dust and chips

Salient characteristics:

General The orthopedic cast cutter shall consist of an electric motor within a metal or plastic housing, an on-off line switch, and a line cord The motor by means of the incorporated mechanism shall impart non-rotating, high frequency, oscillating motion to the attached circular saw blade

The orthopedic vacuum cleaner shall be a compact unit mounted on wheels or casters, and shall be easily emptied The housing shall have a handle on top or integrated with the stand assembly The unit shall include an electric motor-driven vacuum source, an on-off switch on the housing, and shall have a flexible plastic vacuum intake tubing whose proximal end attaches to the housing with a twist lock, snap latch, or other secure mechanism

Material All the plastic components of both the cast cutter and vacuum cleaner shall be fabricated from materials fully capable of performing their intended function All metal components shall be fabricated from suitable corrosion-resistant metal processed as they can perform their intended function

Style, design, and dimensions The cast cutter shall be a non-rotating, high frequency, oscillating circular saw The motor shall operate on either 110 or 230 volts, 50 or 60 Hertz (Hz) Should this motor or the vacuum cleaner motor operate on only 230 volt 50/60 Hz (or 230 volt 50/60 Hz only), then an external step-up (or step-down) transformer must be supplied with the unit

AMSC N/A FSC 65.5

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The motor shall be fan-cooled, permanently lubricated universal type, rated at 0 1 horsepower or more at 10,000 revolutions per minute (RPM) continuous Motor brush holders shall be accessible through brush covers

The cast cutter shall conform to the applicable requirements of Underwriters Laboratories (UL) Standards for Safety for Electric Tools 544 A label or listing of UL shall be acceptable as proof of the equipment conforms to the above A copy of each report shall be furnished to the contracting officer (See Note 2) The power cord shall be a minimum of 9 feet long, 3-wire construction with appropriate 3-prong plug for operation with a vacuum cleaner having no power cord in or clipped on to the vacuum cleaner hose For vacuum cleaners having a power cord in or clipped on to the vacuum cleaner hose, the cast cutter cord shall be one foot long and have a 3-prong plug compatible with that hose cord In either case, the power cord shall be provided with either a spring-type component or an integrally molded component to provide cable strain relief of the housing end Oscillatory action shall be via a cam and open link mechanism, mechanically fastened to the offset blade arbor Gear transmission design is not acceptable The housing shall be compact, rigid, and corrosion-resistant and fitted adequate gasketing to prevent oil leakage The cleaner shall be in accordance with DPSC Drawing 24160

The vacuum cleaner shall be 15 plus or minus 2 inches in diameter (including wheels) and 20 plus or minus 2 inches high (excluding the 20 inch cast cutter pole), or it shall be 9 inches in diameter and 20 inches high (20 inches diameter and 40 inches high including the stand) The 1 7 to 2 horsepower motor shall be fan-cooled, permanently lubricated, sealed ball bearings capable of operating on 110 or 230 volts, 50/60 Hz The vacuum hose shall be 1-1/4 plus or minus 1/4 inch ID and a minimum of 4-1/2 feet long The power cord for the cast cutter which connects the cast-cutter to the vacuum cleaner shall be located either in the hose or fastened outside at 15 inch intervals The plug on this cord and on the cast cutter shall be compatible The proximal end of the hose shall attach to the vacuum cleaner inlet port with a twist-lock, snap latch, or other secure mechanism while the distal end shall have an arbor assembly which attaches over the cast cutter blade (See Note 3) The vacuum cleaner power cord shall be at least 6 feet long

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Performance The cast cutter including power cord and plug and 2-1/2 inch blade but excluding carrying case, 2 inch blade, and wrench, shall weigh 5 pounds or less All screws, pins, bolts, and similar parts shall be installed so they won't loosen during use

With the power cord supported or detached, the center of balance of the cast cutter shall be 6-5/16 +/- 1/2 inches from the front side of the mounted 2-1/2 inch blade measured parallel to the longitudinal axis of the housing)

When operated from a 110-120 volt, 60 Hertz power source, the cast cutter no-load oscillations per minute shall be 17,000 minimum The cast cutter full-load oscillation per minute shall be 13,000 minimum The no-load current shall be 1 2 amperes maximum The maximum motor housing temperature rise after 5 minutes of full load operation shall be 40 degrees Fahrenheit above ambient when starting cold The cast cutter shall be capable of continuous operation without breakdown or failure under full load for 100 hours minimum The switch shall function reliably and the unit shall be free of excessive noise and vibration

The vacuum cleaner shall be capable of generating a minimum of 46 inches of water suction There shall be no air leaks at the tank gasket or hose connection At this time the power draw shall not exceed that required for 2 horsepower operation (1660 watts) The switch shall function reliably and the unit shall be free of excessive noise and vibration When operated at 110-120 volt, 60 Hertz, at a full load power the maximum temperature rise of the motor housing shall be 75 degrees Fahrenheit above ambient after 5 minutes of operation when started cold The motor shall be capable of 100 hours continuous operation Once a cast cutter or vacuum cleaner unit has passed this test, a contractor need not test the unit again unless there has been a major design change in the motor and drive mechanism

Accessories Each cast cutter shall be supplied with one 2 inch and one 2-1/2 inch blade (See Note 4) The cast cutter shall also be supplied with a wrench as shown on Drawing 24160 Each vacuum cleaner or pole shall be supplied with a wall mounting bracket

Each vacuum cleaner shall be supplied with a semi-disposable cloth/plastic bag or filter and all other components necessary to perform its function



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Each cast cutter and each vacuum cleaner shall be supplied with 2 sets of instructions for use Alternatively, each cast cutter/vacuum cleaner combination shall be supplied with 2 sets of instructions which covers both the cast cutter and vacuum cleaner

The instructions shall include Table of Contents, General Information, Operation, Equipment (if any are required), Accessories, Consumable Item (if any), Parts List, Drawings, and Storage

Provisioning The cast cutter and vacuum cleaner shall be as specified in the solicitation or contract There are no requirements for a reusable container for this item

Identification marking The cast cutter and vacuum cleaner shall be each marked on a plate or permanent label with the name or registered trademark of the manufacture, their address type or model number, serial number, voltage, full load current, and cycles to be used for operation

Workmanship Both the cast cutter and vacuum cleaner shall be free from burrs, sharp edges, bare electrical wiring, and other defects which detract from its appearance or impair its serviceability

Visual and dimensional examination The cast cutter and vacuum cleaner shall be examined to verify that design, construction and workmanship are in accordance with applicable requirements

Materials The supplier shall be responsible for performing tests to assure that materials are in accordance with requirements

Unit Each (EA) One cast cutter and vacuum cleaner, complete with all accessories and manuals as specified, constitutes one unit

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

Records Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate

Inspection Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory applicants or procedures) and testing (determination by technical means and chemical properties) of the item

Sampling for examination Sampling for examination shall be conducted in accordance with MIL-STD-105, and as specified herein Unit of product for examination purposes shall be one forceps Sampling shall be Inspection level II with an acceptable quality level (AQL) of 1 0

Examination Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein Unit of product for test purposes shall be one cast cutter and vacuum cleaner Sampling shall be Inspection level S-2 with an AQL of 1 0

Tests Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more text characteristics

Contractor certification The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract



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Metric products Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the lastest revision of Federal Standard No 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the products is acceptable

The contracting officer has the option of accepting or rejecting the product

Regulatory requirements

Federal Food, Drug, and Cosmetic Act If the product covered by this document has been determined by the U S Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug Cosmetic Act, as amended, and regulations promulgated thereunder In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal State, and local statutes, ordinances, and regulations

Recovered materials The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 as amended, and as implemented by Federal Acquisition Regulation Subpart 23 4 to the maximum extent practical

Preservation, packaging, packing, labeling, and marking Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc , involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and /or order



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NOTE 1 The following National Stock Number is covered by this document

| NSN | ITEM IDENTIFICATION |
|---|---|
| 6515-01-304-1003 | Cutter-Vacuum, Orthopedic Cast, 120/230 Volt, 50-60 Hertz, AC |

NOTE 2: In the lieu of the label or listing by the Underwriters' Laboratories Inc , the contractor may submit a laboratory report by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the assurance of a certificate The contractor shall submit the laboratory report which shall include test results and methods used to determine compliance, to the contracting officer for approval The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability

NOTE 3: Should manufacturer cast cutter be used with another manufacturer's vacuum cleaner, an adapter for that specific combination is necessary The requirements for this situation are not covered by this commercial item description

NOTE 4: The two inch blade is covered by National Stock Number (NSN) 6515-00-323-4515 The 2-1/2 inch blade is covered by NSN 6515-00-323-4520.

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order)

| MILITARY INTERESTS: | PREPARING ACTIVITY |
|---|---|
| Custodians | DoD-MR |
| Army-MD | Agent |
| Navy-MS | |
| Air Force-03 | DLA-DM |



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CIVIL AGENCY COORDINATING ACTIVITIES

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4765



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*NOTE.* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| A-A-54162 | CUTTER-VACUUM, ORTHOPEDIC CAST (110/230 V, 50/60 Hz, AC) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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